

Trainings ▾

Preparatory Steps

- Remove the air piping from the intake manifold. Refer to the OEM service manual.
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html)
- Remove the engine brakes, if applicable. Refer to Procedure 020-024 in Section 20.
(/qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html)



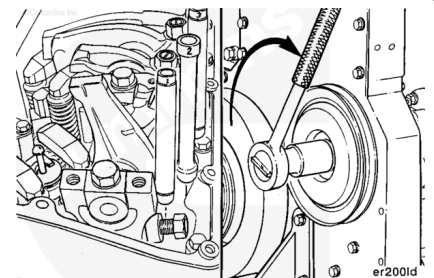
Remove

Mark the position of the valve push tubes and injector push rods as they are removed. The push tubes and push rods **must** be installed in the same positions when the engine is assembled.

Loosen the rocker lever adjusting screws, and remove the valve push tubes and injector push rods.

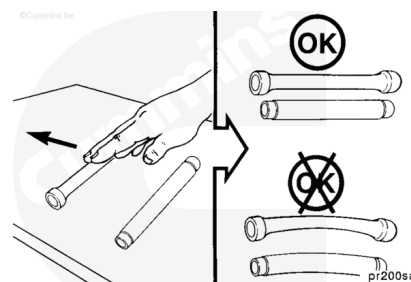
Hold the push tubes and push rods while loosening the adjusting screws. Do **not** let them fall into the engine while rotating the accessory drive pulley.

Note : Some push tubes and rods are under compression because the valves are open. Rotate the crankshaft **clockwise** with the accessory drive pulley to relieve the spring tension.



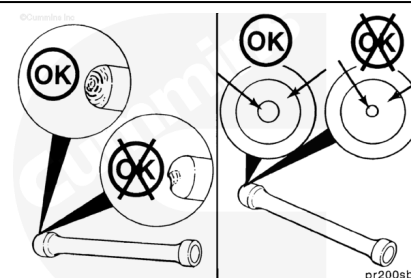
Inspect for Reuse

Inspect the straightness of the push tubes and push rods by rolling them on a level bench. If any push tube or push rod is bent, it **must** be replaced.



Inspect the ball-and-socket ends for uneven wear or scratches.

Check the push tubes for oil. If the push tubes are filled with oil, they **must** be replaced.



Inspect the seating pattern in the socket for excessive wear.

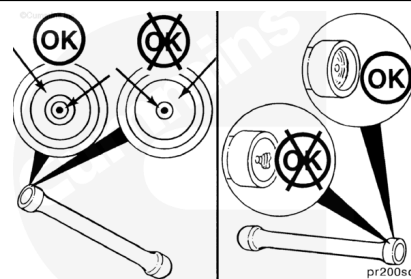
Inspect the ball end for excessive wear.

If excessive wear is found on the ball end, the cam follower sockets **must** be inspected.

Refer to Procedure 004-001 in Section 4.

(</qs3/pubsys2/xml/en/procedures/35/35-004-001-tr.html>)

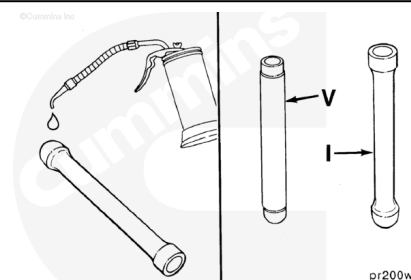
Replace all parts that do **not** meet the reuse guidelines.



Install

Use clean 15W-40 engine oil to coat the ball end of the push tubes and push rods.

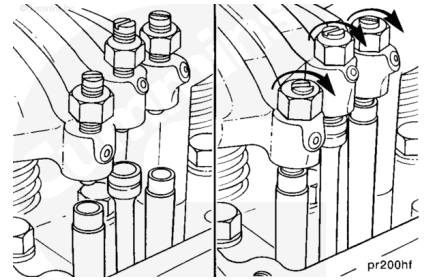
Note : The injector push rods (I) are smaller in diameter and longer than the valve push tubes (V).



Install the valve push tubes and injector push rods.

Position the push tubes and push rods under the rocker lever adjusting screws.

Tighten the adjusting screws enough to hold the push tubes and push rods in position.



Finishing Steps

- Adjust the valves and injectors. Refer to Procedure 003-004 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-004-tr.html>)
- Install the engine brakes, if applicable. Refer to Procedure 020-024 in Section 20. (</qs3/pubsys2/xml/en/procedures/35/35-020-024-tr.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/35/35-003-011-tr.html>)
- Install the air piping to the intake manifold. Refer to the manufacturer's specifications for the correct torque.

